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Lecture 2

Proof of uniqueness part of the fundamental theorem

Arc length s, curvature, and torsion are invariant under rigid motions. For instance if $M: \mathbb{R}^3 \to \mathbb{R}^3$ is a rigid motion and $\alpha = \alpha(t)$ is a parametrized curve.

Then $M(x) = \rho(x) + c \; ; \forall x \in \mathbb{R}^3$

$\therefore (M \circ \alpha)(t) = \rho(\alpha(t)) + c \; \ldots\ldots (1)$

so that $(M \circ \alpha)'(t) = \rho\alpha'(t) \; \ldots\ldots (2)$

Then $|(M \circ \alpha)'(t)|^2 = |\rho(\alpha'(t))|^2$

$= |\alpha'(t)|^2$; [$\because$ orthogonal linear transformation preserves the length of a vector]

Hence $\int_a^b |\alpha'(t)|^2 dt = \int_a^b |(M \circ \alpha)'(t)|^2 dt \; \ldots\ldots (3)$

- **Invariance under rigid motions**

Let $M(x) = \rho(x) + c$, where ρ is an orthogonal linear transformation of determinant +1 (often called rotation) and $c \in \mathbb{R}^3$ constant.

We shall first prove that the rigid motion $M: \mathbb{R}^3 \to \mathbb{R}^3$ preserves arc length, curvature, and torsion.

- **Two properties of a rotation**

Because ρ is orthogonal

$\langle \rho(u), \rho(v) \rangle = \langle u, v \rangle \; \ldots\ldots (1)$

In particular, $|\rho(u)| = |u| \; \ldots\ldots (2)$

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We also need $\rho(u \wedge v) = \rho(u) \wedge \rho(v)$ ..... (3)

Now, $\langle \rho(u) \wedge \rho(v), \rho(w) \rangle = \det(\rho(u), \rho(v), \rho(w))$

$= \det \rho \det(u, v, w)$

$= \det(u, v, w)$: [$\because \det \rho = 1$]

..... (4)

On the other hand $\langle \rho(u \wedge v), \rho(w) \rangle = \langle u \wedge v, w \rangle$: [By (1)]

$= \det(u, v, w)$

..... (5)

Thus the 2 vectors $\rho(u) \wedge \rho(v)$ and $\rho(u \wedge v)$ have the same inner product with every vector $\rho(w)$. Since ρ is surjective, they have the same inner product with every vector in $\mathbb{R}^3$, proving (3).

Arc length

First take any regular parametrized curve $\gamma : J \to \mathbb{R}^3$ with parameter t, and put

$$\hat{\gamma}(t) = M(\gamma(t)) = \rho(\gamma(t)) + c \quad \text{..... (6)}$$

Since ρ is linear and c is constant,

$$\frac{d\hat{\gamma}}{dt} = \rho\left(\frac{d\gamma}{dt}\right)$$

Then by (2), one has $\left|\frac{d\hat{\gamma}}{dt}\right| = \left|\frac{d\gamma}{dt}\right|$ ..... (7)

Integrating over any $[a, b] \subset J$, one has

$$\int_a^b \left|\frac{d\hat{\gamma}}{dt}\right| dt = \int_a^b \left|\frac{d\gamma}{dt}\right| dt \quad \text{..... (8)}$$

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Moreover, if we choose the same base parameter to, the arc-length functions agree:

$$\hat{s}(t) = \int_{t_0}^{t} \left|\frac{d\hat{\gamma}}{d\tilde{u}}\right| d\tilde{u} = \int_{t_0}^{t} \left|\frac{d\gamma}{du}\right| du = s(t) \quad \text{..... (9)}$$

Now, since the cureves are regular.

$$s'(t) = |\dot{\gamma}'(t)| > 0, \quad \text{..... (10)}$$

so that $s(t)$ has inverse $t = t(s)$. Because $\hat{s}(t)$ is the same function as $s(t)$, their inverses also agree: $\hat{t}(s) = t(s)$ ..... (11)

Now, define The arc-length parametizations by

$$\Gamma(s) = \gamma \circ t(s) = \gamma(t(s)), \quad \hat{\Gamma}(s) = \hat{\gamma}(\hat{t}(s)) \quad \text{..... (12)}$$

Then: $\hat{\Gamma}(s) = \hat{\gamma}(\hat{t}(s))$

$= \hat{\gamma}(t(s))$: [By (11)]

$= M(\gamma(t(s)))$: [By (6)]

$= M(\Gamma(s))$: [By (12)] ..... (12)

If we abuse notations and ~~r~~ use the symbols $\gamma, \hat{\gamma}$ for their arc length parametrizations instead of Γ and $\hat{\Gamma}$, we have

$$\hat{\gamma}(s) = M(\gamma(s)) \quad \text{..... (13)}$$

Now, we check curvature and torsion invariance using unit speed/arc length parametrizations.

Curvature

Taking the derivative of (13),

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$$\hat{\gamma}'(s) = \rho(\gamma'(s)) \quad \ldots\ldots\ldots (14)$$

$\gamma(s)$ is of unit speed, i.e., $|\gamma'(s)| = 1$. In other words, $\gamma'(s)$ is a vector of unit length so that the rotated vector $\rho(\gamma'(s)) = \hat{\gamma}'(s)$ will also be of unit length i.e., $\hat{\gamma}(s)$ is also of unit speed.

Differentiating (14) once more,

$$\hat{\gamma}''(s) = \rho(\gamma''(s)) \quad \ldots\ldots\ldots (15)$$

Therefore, $k_{\hat{\gamma}}(s) = |\hat{\gamma}''(s)| = |\rho(\gamma''(s))| \; ; [\text{by } (15)]$

$$= |\gamma''(s)| \; ; [\text{by } (2)]$$

$$= k_\gamma(s) \quad \ldots\ldots\ldots (16)$$

The Frenet frame

Assume $k_\gamma > 0$. The tangent vectors satisfy (we're using $\{T, N, B\}$ as Frenet frame)

$$\hat{T}(s) = \hat{\gamma}'(s) = \rho(\gamma'(s)) \; ; [\text{By } (14)]$$

$$= \rho(T(s)) \quad \ldots\ldots\ldots (17)$$

Using curvature invariance (16),

$$\hat{N}(s) = \frac{\hat{\gamma}''(s)}{k_{\hat{\gamma}}(s)} = \frac{\rho(\gamma''(s))}{k_\gamma(s)} \; ; [\text{using } (15) \text{ and } (16)]$$

$$= \rho\left(\frac{\gamma''(s)}{k_\gamma(s)}\right) = \rho(N(s)) \quad \ldots\ldots (18)$$

Finally, $\hat{B}(s) = \hat{T}(s) \wedge \hat{N}(s) = \rho(T(s)) \wedge \rho(N(s)) \; ; [\text{By } (17) \text{ and } (18)]$

$$= \rho(T(s) \wedge N(s)) \; ; [\text{By } (3)]$$

$$= \rho(B(s)) \quad \ldots\ldots\ldots (19)$$

Thus, we have,

$$\boxed{\begin{aligned}\hat{T}(s) &= \rho(T(s)) \\ \hat{N}(s) &= \rho(N(s)) \\ \hat{B}(s) &= \rho(B(s))\end{aligned}} \quad \text{.....(20)}$$

Torsion

Differentiate the last of the boxed set of equations given by (20):

$$\hat{B}'(s) = \rho(B'(s)) \quad \text{.....(21)}$$

since $B'(s) = \tau(s)N(s)$, one has

$$\hat{B}'(s) = \rho(\tau_\gamma(s)N(s))$$

$$= \tau_\gamma(s)\rho(N(s))$$

$$= \tau_\gamma(s)\hat{N}(s) \text{ ; [By the second of the boxed set of equations given by (20)]} \quad \text{.....(22)}$$

But torsion of $\hat{\gamma}(s)$ is defined by

$$\hat{B}'(s) = \tau_{\hat{\gamma}}(s)\hat{N}(s) \quad \text{.....(23)}$$

Comparing (22) and (23), one gets

$$\tau_\gamma(s) = \tau_{\hat{\gamma}}(s) \quad \text{.....(24)}$$

Align the initial points and Frenet frames

Now, let $\alpha, \bar{\alpha} : I \to \mathbb{R}^3$ be two arc-parametrized curves having the same curvature and torsion:

$$k_\alpha(s) = k_{\bar{\alpha}}(s) = k(s) > 0, \quad \tau_\alpha(s) = \tau_{\bar{\alpha}}(s) = \tau(s) \quad \text{.....(25)}$$

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Fix $s_0 \in I$. Write their initial Frenet frames as $\{T_0, N_0, B_0\}$ and $\{\bar{T}_0, \bar{N}_0, \bar{B}_0\}$. We first justify that a rotation can carry the second frame to the first.

Form the matrices whose columns are these vectors:

$$F_0 = [T_0 \; N_0 \; B_0] \;, \quad \bar{F}_0 = [\bar{T}_0 \; \bar{N}_0 \; \bar{B}_0] \quad \cdots\cdots\cdots (26)$$

at the fixed parameter value s_0.

$$T_0 = T(s_0) = \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix}, \; N_0 = N(s_0) = \begin{bmatrix} N_1 \\ N_2 \\ N_3 \end{bmatrix}, \; B_0 = B(s_0) = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix} \quad \cdots\cdots (27)$$

An interlude: (On orientation of an ordered basis)

2 ordered bases (a, b, c) and (a', b', c') of $\mathbb{R}^3$ have the same orientations if the change of basis matrix from (a, b, c) to (a', b', c') has positive determinant.

We have $F_0 = [T_0, N_0, B_0]$ and the standard coordinate frame $\mathcal{E} = [e_1, e_2, e_3]$

F_0: Frenet frame

Express the Frenet vectors in the standard basis:

$$T_0 = \sum_{i=1}^{3} T_i e_i \;, \quad N_0 = \sum_{i=1}^{3} N_i e_i \;, \quad B_0 = \sum_{i=1}^{3} B_i e_i \quad \cdots\cdots\cdots (28)$$

Then

$$F_0 = \begin{bmatrix} T_1 & N_1 & B_1 \\ T_2 & N_2 & B_2 \\ T_3 & N_3 & B_3 \end{bmatrix}, \quad \cdots\cdots\cdots (29)$$

so that the columns of F_0 are the standard coordinates of the Frenet basis vectors.

Now, take $v \in \mathbb{R}^3$, and suppose $v = x_1 T_0 + x_2 N_0 + x_3 B_0 \cdots (29)$

Its Frenet coordinates are

$$[v]_{F_0} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \cdots\cdots (30)$$

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Then $v = x_1(T_1 e_1 + T_2 e_2 + T_3 e_3) + x_2(N_1 e_1 + N_2 e_2 + N_3 e_3)$
$+ x_3(B_1 e_1 + B_2 e_2 + B_3 e_3)$
; [substituting (28) into (29)]

$= (x_1 T_1 + x_2 N_1 + x_3 B_1) e_1 + (x_1 T_2 + x_2 N_2 + x_3 B_2) e_2$
$+ (x_1 T_3 + x_2 N_3 + x_3 B_3) e_3$

So that $[v]_{\mathcal{E}} = \begin{bmatrix} x_1 T_1 + x_2 N_1 + x_3 B_1 \\ x_1 T_2 + x_2 N_2 + x_3 B_2 \\ x_1 T_3 + x_2 N_3 + x_3 B_3 \end{bmatrix}$

$= \begin{bmatrix} T_1 & N_1 & B_1 \\ T_2 & N_2 & B_2 \\ T_3 & N_3 & B_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$= F_0 [v]_{F_0}$

$\therefore [v]_{\mathcal{E}} = F_0 [v]_{F_0}$ ..... (31)

In other words, F_0 takes coordinates relative to F_0 to coordinates relative to $\mathcal{E}$.

The Frenet frame is orthonormal: $|T_0| = |N_0| = |B_0| = 1$

and $\langle T_0, N_0 \rangle = \langle T_0, B_0 \rangle = \langle N_0, B_0 \rangle = 0$

$\therefore F_0^T F_0 = \begin{bmatrix} T_1 & T_2 & T_3 \\ N_1 & N_2 & N_3 \\ B_1 & B_2 & B_3 \end{bmatrix} \begin{bmatrix} T_1 & N_1 & B_1 \\ T_2 & N_2 & B_2 \\ T_3 & N_3 & B_3 \end{bmatrix} = \begin{bmatrix} |T_0|^2 & \langle T_0, N_0 \rangle & \langle T_0, B_0 \rangle \\ \langle N_0, T_0 \rangle & |N_0|^2 & \langle N_0, B_0 \rangle \\ \langle B_0, T_0 \rangle & \langle B_0, N_0 \rangle & |B_0|^2 \end{bmatrix}$

$= I_3$ ..... (32)

Now, calculate $\det F_0 = \det(T_0, N_0, B_0) = \langle T_0 \wedge N_0, B_0 \rangle$
$= \langle B_0, B_0 \rangle = 1 > 0$

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Therefore, F_0 has the same orientation as that of $\mathcal{E}$.

Similarly, for $\bar{F}_0$, $\bar{F}_0\bar{F}_0^T = \bar{F}_0^T\bar{F}_0 = I_3$. with $\det \bar{F}_0 = 1$

Define ρ to have matrix form $R = F_0\bar{F}_0^T$ ..... (33)

Then $R^TR = \bar{F}_0F_0^TF_0\bar{F}_0^T = \bar{F}_0\bar{F}_0^T = I_3$,

and $\det R = \det F_0 \det \bar{F}_0^T = 1$.

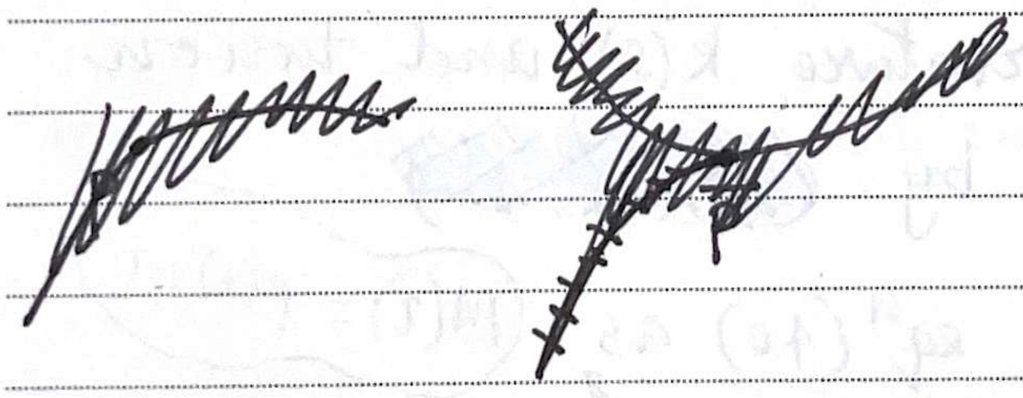
Thus ρ is a rotation. Moreover, from (33), one has

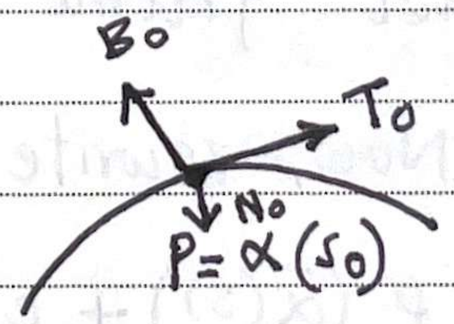
$$R(\bar{F}_0^T)^{-1} = F_0 \Rightarrow R(\bar{F}_0^{-1})^T = F_0$$

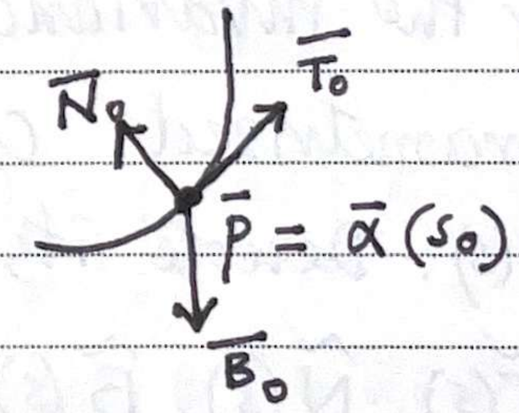
$$\Rightarrow R(\bar{F}_0^T)^T = F_0 \Rightarrow R\bar{F}_0 = F_0, \quad \text{.....(34)}$$

which says precisely that

$$\boxed{\begin{aligned} \rho(\bar{T}_0) &= T_0 \\ \rho(\bar{N}_0) &= N_0 \\ \rho(\bar{B}_0) &= B_0 \end{aligned}} \quad \text{.....(35)}$$







p and $\bar{p}$ are initial points. The rigid motion $M: \mathbb{R}^3 \to \mathbb{R}^3$ is the transformation (in our case)

$$M(\bar{p}) = p \leftarrow \text{see below}$$

Remark. Rotating in $\mathbb{R}^3$ about $\bar{p} \in \mathbb{R}^3$ is given by

$$R_{\bar{p}}(x) = \bar{p} + \rho(x - \bar{p}) \quad \text{.....(36)}$$

when rotation happens w.r.t origin, we have,

$$R_0(x) = \rho(x)$$

Perform the in-place rotation at $\bar{p}$, and then translate by

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$d = \bar{p} - \bar{p}$. The complete motion is

$$M(x) = R_{\bar{p}}(x) + d$$

$$= \bar{p} + \rho(x - \bar{p}) + \bar{p} - \bar{p}$$

$$= \rho(x) - \rho(\bar{p}) + \bar{p} \quad \text{---------- (37)}$$

$M(x) = \rho(x) + c$ with $c = \bar{p} - \rho(\bar{p})$ a constant

Plug in $x = \bar{p}$ to obtain $M(\bar{p}) = \bar{p}$ ---------- (38)

which translates to $M(\bar{\alpha}(s_0)) = \alpha(s_0)$ ---------- (39)

A note: $R_{\bar{p}}(\bar{p}) = \bar{p}$ as rotation about $\bar{p}$ keeps the vector $\bar{p}$ fixed. But $\rho(\bar{p})$ is the rotated vector after one rotates the vector $\bar{p}$ about the origin and hence $\rho(\bar{p}) \neq \bar{p}$.

But indeed, $\rho(0) = 0$.

Now, introduce $\boxed{\beta(s) = M(\bar{\alpha}(s))}$ ---------- (40)

By the invariance results proved earlier, the arc parametrized curve $\beta(s)$ has curvature $k(s)$ and torsion $\tau(s)$. Denote its Frenet frame by $\{\tilde{T}(s), \tilde{N}(s), \tilde{B}(s)\}$. Now, rewrite eq^n (40) as $\beta(s) = \rho(\bar{\alpha}(s)) + c$, where c is a constant ($M(x) = \rho(x) + c$)

Then $\beta'(s) = \rho(\bar{\alpha}'(s))$ ---------- (41)

$$\Rightarrow \tilde{T}(s) = \rho(\bar{T}(s)) \quad \text{---------- (42)}$$

But we already know that $\rho(\bar{T}(s_0)) = T(s_0)$; [By (35)] ..... (43)

And taking $s = s_0$ in (42), one has, $\tilde{T}(s_0) = \rho(\bar{T}(s_0))$ ..... (44)

Combining (43) and (44), one obtains

$$\tilde{T}(s_0) = T(s_0) \quad \text{---------- (45)}$$

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Now, differentiating (41) once again;

$$\beta''(s) = \rho(\bar{\alpha}''(s)) \quad \ldots\ldots\ldots (46)$$

Orthogonality of ρ yields $|\beta''(s)| = |\rho(\alpha''(s))| = |\alpha''(s)|$

$$= k(s) > 0. \quad \ldots\ldots\ldots (47)$$

Therefore, $\tilde{N}(s_0) = \dfrac{\beta''(s_0)}{|\beta''(s_0)|}$

$$= \frac{\rho(\bar{\alpha}''(s_0))}{k(s_0)} \; ; [\text{using (46) \& (47)}]$$

$$= \rho\left(\frac{\bar{\alpha}''(s_0)}{k(s_0)}\right)$$

$$= \rho\left(\bar{N}(s_0)\right) = N(s_0) \quad \ldots\ldots\ldots (48)$$

$$; [\text{By (35)}]$$

Now, using (45) and (48), one obtains the following:

$$\tilde{B}(s_0) = \tilde{T}(s_0) \wedge \tilde{N}(s_0)$$

$$= T(s_0) \wedge N(s_0) = B(s_0) \quad \ldots\ldots\ldots (49)$$

Recall our rigid motion expression $M(x) = \rho_\bullet(x) + c$

with $c = p - \rho(\bar{p})$

where $p = \alpha(s_0)$ & $\bar{p} = \bar{\alpha}(s_0)$

so that $c = \alpha(s_0) - \rho(\bar{\alpha}(s_0)) \quad \ldots\ldots\ldots (50)$

Consequently, $\beta(s_0) = \rho\left(\bar{\alpha}(s_0)\right) + c$

$$= \rho\left(\bar{\alpha}(s_0)\right) + \alpha(s_0) - \rho\left(\bar{\alpha}(s_0)\right) \; ; [\text{using (50)}]$$

$$= \alpha(s_0) \quad \ldots\ldots\ldots (51)$$

- Prove that the aligned frames agree everywhere:

The Frenet frames of α and β satisfy the Frenet equations with the same coefficients k, τ:

$$T' = kN, \qquad \tilde{T}' = k\tilde{N}$$

$$N' = -kT - \tau B, \qquad \tilde{N}' = -k\tilde{T} - \tau\tilde{B} \quad \ldots\ldots\ldots (52)$$

$$B' = \tau N, \qquad \tilde{B}' = \tau\tilde{N}$$

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Define the vector differences

$$\mathcal{U} = T - \tilde{T}, \quad V = N - \tilde{N}, \quad W = B - \tilde{B} \quad \cdots\cdots (53)$$

Then, subtracting the Frenet equations in (52) yields

$$\left.\begin{aligned} (T-\tilde{T})' &= T' - \tilde{T}' = k(N-\tilde{N}) \\ (N-\tilde{N})' &= N' - \tilde{N}' = -k(T-\tilde{T}) - \tau(B-\tilde{B}) \\ (B-\tilde{B})' &= B' - \tilde{B}' = \tau(N-\tilde{N}) \end{aligned}\right\} \quad \cdots\cdots (54)$$

Now, the above set of equations can be cast into the following form using $\mathcal{U}, V, W$ and their derivatives:

$$\boxed{\mathcal{U}' = kV, \quad V' = -k\mathcal{U} - \tau W, \quad W' = \tau V} \quad \cdots\cdots (55)$$

Now, consider the non negative function

$$E(s) = \frac{1}{2}\left(|\mathcal{U}(s)|^2 + |V(s)|^2 + |W(s)|^2\right)$$

Now, for the differentiable vector valued function

$$\mathcal{U}(s), \quad \frac{d}{ds}|\mathcal{U}(s)|^2 = \frac{d}{ds}\langle \mathcal{U}, \mathcal{U} \rangle = 2\langle \mathcal{U}, \mathcal{U}' \rangle \quad \cdots\cdots (56)$$

Applying this identity to all 3 terms and adding them,

$$\frac{d}{ds}\left(\frac{1}{2}\left(|\mathcal{U}(s)|^2 + |V(s)|^2 + |W(s)|^2\right)\right) = \langle \mathcal{U}, \mathcal{U}' \rangle + \langle V, V' \rangle + \langle W, W' \rangle \quad \cdots\cdots (56)$$

Substituting (55) in (56), one obtains,

$$\begin{aligned} E'(s) &= \langle \mathcal{U}, kV \rangle + \langle V, -k\mathcal{U} - \tau W \rangle + \langle W, \tau V \rangle \\ &= k\langle \mathcal{U}, V \rangle - k\langle V, \mathcal{U} \rangle - \tau\langle V, W \rangle + \tau\langle W, V \rangle \\ &= 0 \end{aligned}$$

Hence, $E(s)$ is constant on the interval I.

Now, using (53), one has

$$\left.\begin{aligned} \mathcal{U}(s_0) = T(s_0) - \tilde{T}(s_0) = 0 \;;\; [\text{using (45)}] \end{aligned}\right\} \cdots (57)$$

Similarly, $V(s_0) = W(s_0) = 0$.

From (57), one has, $E(s_0) = 0 \cdots\cdots\cdots (58)$

But $E'(s) = 0 \Rightarrow E(s) = 0$ for every $s \in I$ $\cdots\cdots\cdots (59)$

But each summand in $E(s) = \frac{1}{2}\left(|\mathcal{U}(s)|^2 + |V(s)|^2 + |W(s)|^2\right)$ is nonnegative. Their sum vanish implies that each summand vanishes. Thus,

$$\left.\begin{aligned} \mathcal{U}(s) = T(s) - \tilde{T}(s) = 0 \Rightarrow T(s) = \tilde{T}(s) \\ \text{Similarly, } N(s) = \tilde{N}(s),\; B(s) = \tilde{B}(s),\; \forall s \in I \end{aligned}\right\} \cdots (60)$$

- <u>Prove that the aligned curves agree, then undo the rigid motion</u>

We know that $\alpha'(s) = T(s)$ and $\beta'(s) = \tilde{T}(s)$

Then by (60), $(\alpha - \beta)' = 0$.

Therefore, $\alpha - \beta$ is constant. Using (51), one obtains

$$\alpha(s) - \beta(s) = \alpha(s_0) - \beta(s_0) = 0.$$

Hence, $\boxed{\alpha(s) = \beta(s) \;,\; \forall s \in I}$ $\cdots\cdots\cdots (61)$

But recall (40) that $\beta = M_0 \bar{\alpha} \Rightarrow \alpha = M_0 \bar{\alpha}$; [using (61)]

Consequently, [illegible] $\cdots\cdots (62)$

Consequently, $\bar{\alpha} = M_0^{-1} \alpha$. $\cdots\cdots\cdots (63)$

[illegible]. We've seen $\alpha(s) = \beta(s) = \rho(\bar{\alpha}(s)) + c$

Then $\alpha(s) - c = \rho(\bar{\alpha}(s))$ $\cdots\cdots\cdots (64)$

since ρ is a rotation, it is invertible. Apply ρ^{-1} on both sides of (64),

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$$\rho^{-1}(\alpha(s) - c) = \rho^{-1}(\rho(\bar{\alpha}(s))) = \bar{\alpha}(s) \quad \ldots\ldots (65)$$

Finally, ρ^{-1} is linear so that

$$\rho^{-1}(\alpha(s) - c) = \rho^{-1}(\alpha(s)) - \rho^{-1}(c) \quad \ldots\ldots (66)$$

Comparing (65) and (66), one obtains,

$$\bar{\alpha}(s) = \rho^{-1}(\alpha(s)) - \rho^{-1}(c) \quad \ldots\ldots (67)$$

(63) reveals the fact that the 2 curves α and $\bar{\alpha}$ are related by a rigid motion

□

- We will prove the existence part of the Fundamental theorem in the next class.